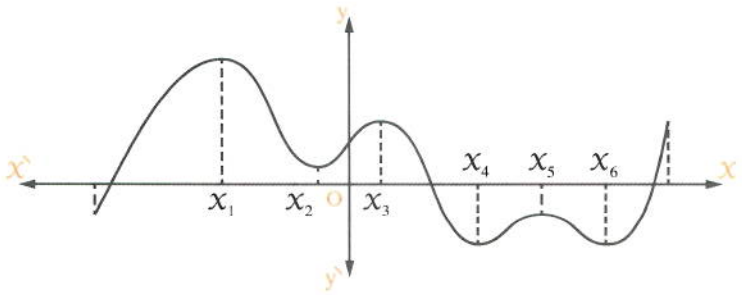
Unit 2

Lesson Two

Local maxima and minima (Local extrema)

⊙ **If the function f is defined on the interval "I" and represented graphically as follow :**



Notice that

1. At $x_1 \in]a , b[$, the value of the function is more than any value just to the right or to the left of x_1 and so , the function has a local maximum value at x_1 and also at x_3 and x_5
2. At $x_2 \in]a , b[$, the value of the function is less than any value just to the right or to the left of x_2 and so, the function has a local minimum value at x_2 and also at x_4 , x_6

Definition

Local maximum and minimum values :

If f is a continuous function whose domain is I and $c \in I$, then the function f has :

1. **A local maximum value at $x = c$** if an open interval is found $]a , b[\subset I$ containing c where $f(x) \leq f(c)$ for each $x \in]a , b[$
2. **A local minimum value at $x = c$** if an open interval is found $]a , b[\subset I$ containing c where $f(x) \geq f(c)$ for each $x \in]a , b[$

We will study some theorems help us to find the local maximum value and the local minimum value.

Theorem " 1 "

If the function f is continuous in the interval $]a , b[$ and if the function f has a local maximum or minimum value at $c \in]a , b[$, then :

- $\dot{f}(c) = 0$ or $\dot{f}(c)$ is not exist

- **From the previous theorem we deduce that :**

To find the local maximum or local minimum value , we have to find the critical points of the function f [the points at which $\dot{f}(x) = 0$ or $\dot{f}(x)$ does not exist] because the maximum value and minimum value exist only at the critical points.

Theorem " 2 " (the first derivative test)

If $(c , f(c))$ is a critical point of the function f which is continuous at c and there is an open interval around c where :

1. $\dot{f}(x) > 0$ at $x < c$, $\dot{f}(x) < 0$ at $x > c$, then $f(c)$ is a local maximum value.
2. $\dot{f}(x) < 0$ at $x < c$, $\dot{f}(x) > 0$ at $x > c$, then $f(c)$ is a local minimum value.
3. If the sign of $\dot{f}(x)$ on both sides of c does not change then the function has no local maximum or local minimum value at c

- **The following figures show that :**



Theorem " 3 " (the second derivative test)

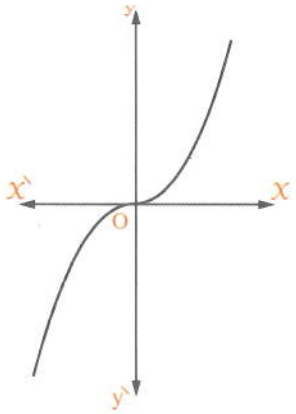
If f is differentiable function twice on an open interval contains c where $\dot{f}(c) = 0$ and

1. $\ddot{f}(c) < 0$, then $f(c)$ is local maximum value.
2. $\ddot{f}(c) > 0$, then $f(c)$ is local minimum value.
3. $\ddot{f}(c) = 0$, then the 2nd derivative test failed to determine the kind of the point $(c , f(c))$ if it is local maximum or local minimum value , in this case we use the first derivative test.

Remarks

1. The converse of theorem (1) is not always true

i.e. If $\dot{f}(x_1) = 0$ of a differentiable function at x_1 that is not necessary means that there is a local maximum or local minimum value at this point.



For example : The function $f(x) = x^3$

$\dot{f}(0)$ = zero and the function has no local maximum or local minimum value at $x = 0$ where the function is increasing on $\mathbb{R}$ (as shown in the figure).

Notice that

The function has a critical point at $x = 0$

2. The local maximum points and local minimum points are critical points

but the converse is not always true.

i.e. (The critical points could be maximum , minimum or neither maximum nor minimum). the following charts illustrate that.

If f (a) is defined and $\dot{f}$ (a) = 0 / f (a) is defined and $\dot{f}$ (a) not exist **then** The point (a , f (a)) is critical **and So it is**

- local maximum If $\dot{f}(x)$ changes its sign around (a) from (+) to (–)
- local minimum If $\dot{f}(x)$ changes its sign around (a) from (–) to (+)
- Neither maximum nor minimum If $\dot{f}(x)$ does not change its sign around (a)

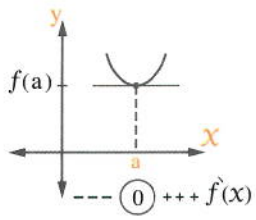
3. If the function f is only increasing (or decreasing) on an interval , so the function has no local maximum or local minimum in this interval.

4. The critical point at which the first derivative = 0

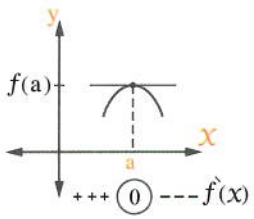
i.e. The tangent is horizontal at this point sometimes is called stationary point.

The following are some figures illustrate the relation between the critical points and the local maximum and local minimum of the function [where f (a) is defined].

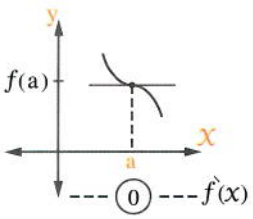
1 **If $f'(a) = 0$ i.e. The tangent is horizontal :**



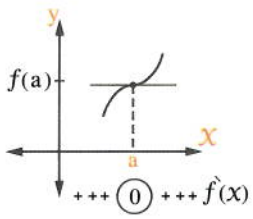
∴ f (a) is a local minimum value



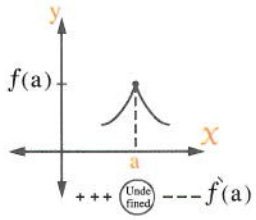
∴ f (a) is a local maximum value



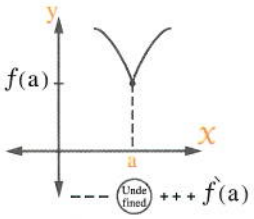
∴ f (a) is neither maximum nor minimum value



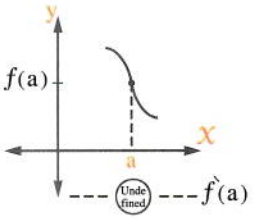
2 **If $f'(a)$ is undefined :**



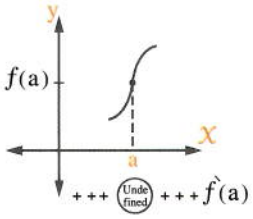
∴ f (a) is a local maximum value



∴ f (a) is a local minimum value



∴ f (a) is neither maximum nor minimum value



Steps to study the existence of the local maximum and local minimum values of continuous function not including the constant function :

1. Determine the domain of the function.
2. Calculate $f'(x)$
3. Find out the critical points (**i.e.** The points at which $f'(x) = 0$ or not exist) , let the x-coordinate of one of them be x_1
4. Determine the type of the critical point if its maximum or minimum by one of the next two methods.

- **First method :**

If

- $f'(x) > 0$ on the left side of x_1 , $f'(x) < 0$ on the right side of x_1 **i.e.** $f'(x)$ changes its sign around x_1 from positive to negative → **then** → $(x_1 , f(x_1))$ is a point of local maximum.
- $f'(x) < 0$ on the left side of x_1 , $f'(x) > 0$ on the right side of x_1 **i.e.** $f'(x)$ changes its sign around x_1 from negative to positive → **then** → $(x_1 , f(x))$ is a point of local minimum.

• **Second method :** By using the sign of $f''(x)$ as follow :

If

- $f'(x_1) = 0$, $f''(x_1) < 0$ (negative) **then** → $(x_1 , f(x_1))$ is a point of local maximum.
- $f'(x_1) = 0$, $f''(x_1) > 0$ (positive) **then** → $(x_1 , f(x_1))$ is a point of local minimum.

Notice that

The 2nd method not suitable to study the local maximum and local minimum value if $f''(x) = 0$

Example 1

Find the local maximum value and local minimum value of the function f : $f(x) = 2x^3 - 3x^2 - 36x + 14$

Solution

$\because f(x) = 2x^3 - 3x^2 - 36x + 14$

$\therefore$ The domain of the function is $\mathbb{R}$

, $f'(x) = 6x^2 - 6x - 36 = 6(x^2 - x - 6)$

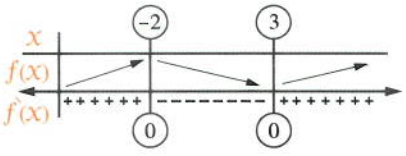
$= 6(x - 3)(x + 2)$

Put $f'(x) = 0$

$\therefore x = -2$ or $x = 3$

$\therefore$ There are critical points at $x = -2$ and $x = 3$

⊙ By using the 1st derivative test , determine the sign of $f'(x)$ as follow :



, $\because$ around $x = -2$, the sign of $f'(x)$, changes from positive to negative

$\therefore$ At $x = -2$ there is a local maximum value , the maximum value = $f(-2) = 58$

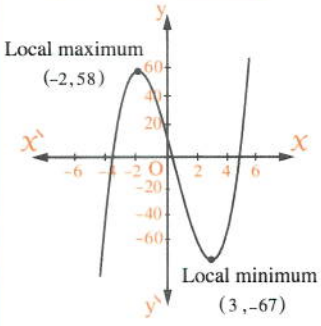
, $\because$ around $x = 3$, the sign of $f'(x)$ changes from negative to positive

$\therefore$ At $x = 3$ there is a local minimum value , the minimum value = $f(3) = -67$

Notice that

The solution using the first derivative test.

For reading only



Example 2

Find the local maximum and local minimum value of the function $f : f(x) = x^4 - 18x^2$

Solution

$\because f(x) = x^4 - 18x^2$ $\quad\quad$ $\therefore$ The domain of the function is $\mathbb{R}$

$, f'(x) = 4x^3 - 36x = 4x(x^2 - 9) = 4x(x-3)(x+3)$

Put $f'(x) = 0$

$\therefore x = 0$ or $x = 3$ or $x = -3$

$, \because f''(x) = 12x^2 - 36$ $\quad\quad$ $\therefore f''(0) = -36$ (negative)

$\therefore$ at $x = 0$, the function has local maximum value which is $f(0) = 0$

$, f''(3) = 72$ (positive)

$\therefore$ at $x = 3$, the function has local minimum value which is $f(3) = -81$

$, f''(-3) = 72$ (positive)

$\therefore$ at $x = -3$, the function has local minimum value which is $f(-3) = -81$

Notice that

The solution using the second derivative test.

Example 3

Find the local maximum and local minimum value of the function $f : f(x) = x^4 + 1$

Solution

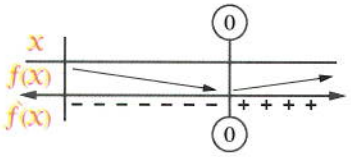
$\because f(x) = x^4 + 1$ $\quad\quad$ $\therefore$ The domain of the function is $\mathbb{R}$

$, f'(x) = 4x^3$ $\quad\quad$ $, f''(x) = 12x^2$

put $f'(x) = 0$ $\quad\quad$ $\therefore x = 0$

$, f''(0) = 12(0)^2 = 0$

$\therefore$ This test is not suitable to determine the type of the critical point , so study the sign of $f'(x)$ around $x = 0$ (first derivative test) as follow



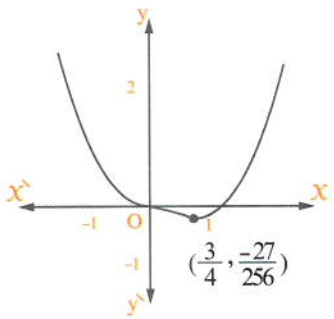
$\therefore$ At $x = 0$ there is a local minimum value of the function which is $f(0) = 1$

Example 4

Find the local maximum value and local minimum value of the function
$f : f(x) = x^4 - x^3$

Solution

For reading only



$\because f(x) = x^4 - x^3$

$\therefore$ The domain of the function is $\mathbb{R}$

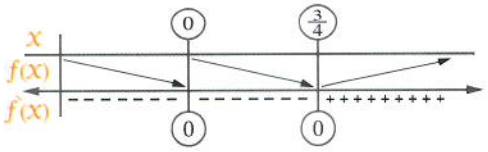
, $f'(x) = 4x^3 - 3x^2 = x^2(4x - 3)$

, put $f'(x) = 0$

$\therefore x = 0$ or $x = \frac{3}{4}$

$\therefore$ There is a critical point at $x = 0$ and at $x = \frac{3}{4}$

around $x = 0$, the sign of $f'(x)$ does not change



$\therefore$ At $x = 0$ there is no local maximum or minimum value

, around $x = \frac{3}{4}$ the sign of $f'(x)$ changes from negative to positive

$\therefore$ At $x = \frac{3}{4}$ there is a local minimum value which is $f\left(\frac{3}{4}\right) = \frac{-27}{256}$

Example 5

Find the local maximum and local minimum value of the function : $y = \frac{1 - x^2}{1 + x^2}$

Solution

$\because y = \frac{1 - x^2}{1 + x^2}$ $\quad\therefore \frac{dy}{dx} = \frac{-2x(1 + x^2) - 2x(1 - x^2)}{(1 + x^2)^2} = \frac{-4x}{(1 + x^2)^2}$

, $\frac{d^2y}{dx^2} = \frac{-4(1 + x^2)^2 + 4x \times 2(1 + x^2) \times 2x}{(1 + x^2)^4}$

$\therefore \frac{dy}{dx} = 0$ at $-4x = 0$ so $x = 0$

, $\left[\frac{d^2y}{dx^2}\right]_{x=0} = \frac{-4 + 0}{1} = -4 < 0$

$\therefore$ At $x = 0$ there is a local maximum value , which is $f(0) = 1$

Important note

When $f(x)$ is a fractional function - as the previous example - let $f'(x) = \frac{h(x)}{g(x)}$ and $f'(x) = 0$ at $x = x_1$ **i.e.** $h(x_1) = 0$, $g(x)$ is a positive quantity

, $\because f''(x) = \frac{g(x)\, h'(x) - h(x)\, g'(x)}{[g(x)]^2}$

and to determine the type of the point at x_1 study the sign of $f''(x_1)$

$f''(x_1) = \frac{g(x_1)\, h'(x_1) - h(x_1)\, g'(x_1)}{[g(x_1)]^2}$

$\because h(x_1) = 0$, $[g(x_1)]^2$ is always positive

$\because f''(x_1) = \frac{h'(x) \times [\text{positive}] - 0}{\text{positive quantity}}$

$\therefore$ The sign of $f''(x_1)$ is the same as the sign of the first term in the numerator of $f''(x_1)$ which is $h'(x_1)$ because the second term in the numerator = 0 and the denominator is a positive quantity and so it is sufficient to solve this term to save time and effort.

Example 6

Find the points of local maximum and local minimum of the function

$f : f(x) = \frac{3-x}{x^2-x-2}$ where $x \neq 2$, $x \neq -1$

Solution

$\because f(x) = \frac{3-x}{x^2-x-2}$

$\therefore f'(x) = \frac{-(x^2-x-2)-(2x-1)(3-x)}{(x^2-x-2)^2} = \frac{-x^2+x+2+2x^2-7x+3}{(x^2-x-2)^2}$

$= \frac{x^2-6x+5}{(x^2-x-2)^2} = \frac{(x-1)(x-5)}{(x^2-x-2)^2}$ where $x \neq 2$, $x \neq -1$

, $f''(x) = \frac{(2x-6)(x^2-x-2)^2 - 2(x^2-x-2)(2x-1)(x^2-6x+5)}{(x^2-x-2)^4}$, $x \neq 2$, $x \neq -1$

$\because f'(x) = 0$ at $(x-1)(x-5) = 0$ **i.e.** at $x = 1$ or $x = 5$

$\therefore$ There are critical points at $x = 1$ and $x = 5$

$\therefore f''(1) = \dfrac{(2-6) \times (\text{positive}) - \text{zero}}{\text{positive quantity}} = \text{negative quantity}$

$\therefore$ At $x = 1$ the function has a local maximum value is $f(1) = -1$

$\therefore$ The point which the function has local maximum value is $(1 , -1)$

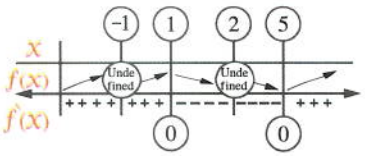
, $f''(5) = \dfrac{(10-6) \times (\text{positive}) - \text{zero}}{\text{positive quantity}} = \text{positive quantity}$

$\therefore$ At $x = 5$ the function has local minimum value is $f(5) = \dfrac{-1}{9}$

$\therefore$ The point which the function has local minimum value is $\left(5 , \dfrac{-1}{9}\right)$

Another Solution (without using 2nd derivative)

Investigate the sign of $f'(x)$ around the critical point as follow



Notice that

It's better to solve the fractional function using investigation of $f'(x)$ sign.

$\therefore$ At $x = 1$ there is a local maximum value.

, At $x = 5$ there is a local minimum value.

Point of local maximum is $(1 , -1)$

Point of local minimum is $\left(5 , -\dfrac{1}{9}\right)$

Example 7

Find the points of local maximum and local minimum of the function

$f : f(x) = x + \dfrac{1}{x}$ where $x \neq$ zero

Solution

$\because f(x) = x + \dfrac{1}{x}$

$\therefore$ The domain of the function $= \mathbb{R} - \{0\}$

, $f'(x) = 1 - \dfrac{1}{x^2}$, $f''(x) = \dfrac{2}{x^3}$

put $f'(x) = 0$

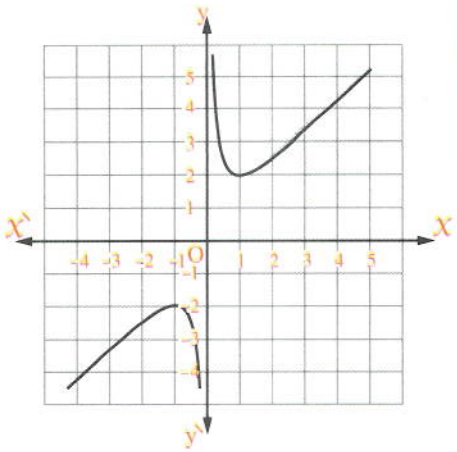
$\therefore 1 - \dfrac{1}{x^2} = 0$ $\therefore x^2 = 1$

$\therefore x = \pm 1$, there are two critical points

at $x = 1$, $x = -1$

, $\because f''(1) = 2 > 0$

For reading only



$\therefore$ There is a local minimum value at $x = 1$ which is $f(1) = 2$

$\therefore$ The point of minimum value is $(1 , 2)$

, $\because f'(-1) = -2 < 0$

$\therefore$ There is a local maximum value at $x = -1$ which is

$f(-1) = -2$

Remark

From the previous example we notice that the local maximum value (-2) < the local minimum value (2) and so we deduce that the local maximum value could be less than the local minimum value

Example 8

Find the points of local maximum and local minimum of the function

$f : f(x) = x^{\frac{1}{3}}$

Solution

$\because f(x) = x^{\frac{1}{3}}$

$\therefore$ The domain of the function is $\mathbb{R}$

$\therefore f'(x) = \frac{1}{3} x^{-\frac{2}{3}} = \frac{1}{3\sqrt[3]{x^2}}$

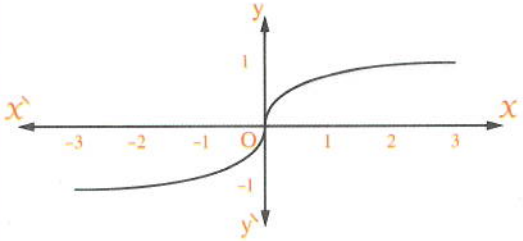
$\therefore f'(x)$ is undefined at $x = 0$

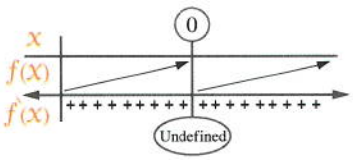
$\therefore$ There is a critical point at $x = 0$

$\therefore$ The sign of $f'(x)$ does not change around $x = 0$

$\therefore$ At $x = 0$ there is no local maximum or local minimum

For reading only





Example 9

Find the local maximum and local minimum values of the function

$f : f(x) = 2x - 3x^{\frac{2}{3}}$

Solution

$\because f(x) = 2x - 3x^{\frac{2}{3}}$

$\therefore$ The domain of the function is $\mathbb{R}$

$f'(x) = 2 - 2x^{-\frac{1}{3}} = \dfrac{2(x^{\frac{1}{3}} - 1)}{x^{\frac{1}{3}}}$

, put $f'(x) = 0$

$\therefore x^{\frac{1}{3}} = 1 \qquad \therefore x = 1$

, $f'(x)$ is undefined at $x = 0$

$\therefore$ There is a critical point at $x = 0$, $x = 1$

$\therefore$ At $x = 0$, the sign of $f'(x)$ changes from positive to negative.

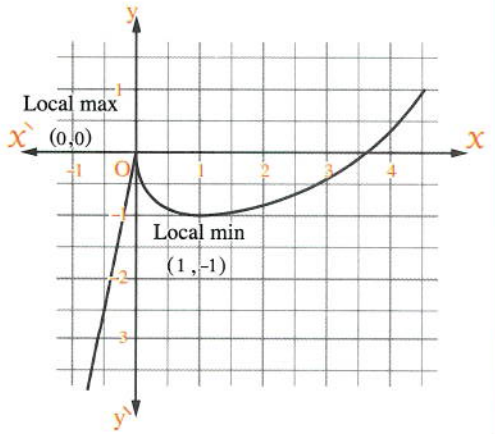
$\therefore$ At $x = 0$ there is a maximum value.

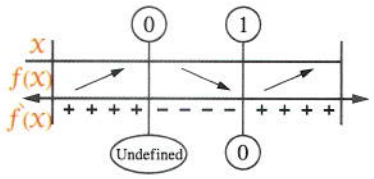
$\therefore$ The local maximum value $= f(0) = 0$

, $\because$ at $x = 1$ the sign of $f'(x)$ changes from negative to positive.

$\therefore$ At $x = 1$ there is a local minimum value which is $f(1) = -1$

For reading only





Example 10

Find the local maximum and local minimum values of the function

$$f : f(x) = \begin{cases} 2x + x^2 , & x \le 0 \\ 2x - x^2 , & x > 0 \end{cases}$$

Solution

$\because$ The function is continuous at $x = 0$, $f(0) = 0$

, $f'(x) = 2 + 2x$ for $x < 0$

, $f'(x) = 2 - 2x$ for $x > 0$

By studying the differentiability at $x = 0$ we find that :

$f'(0^-) = 2$, $f'(0^+) = 2$ $\therefore f'(0) = 2$

$\therefore f'(x) = \begin{cases} 2+2x & , x<0 \\ 2 & , x=0 \\ 2-2x & , x>0 \end{cases}$ $\therefore f'(x)=0$ at $2+2x=0$

$\therefore x=-1$ (verifies that $x<0$) or $2-2x=0$

$\therefore x=1$ (verifies that $x>0$) $\therefore$ There are critical points at $x=-1$, $x=1$

, $\because f''(x)=2$ when $x<0$ $\therefore f''(-1)=2>0$

$\therefore$ At $x=-1$ the function has local minimum value is $f(-1)=-1$

, $\because f''(x)=-2$ when $x>0$ $\therefore f''(1)=-2<0$

$\therefore$ At $x=1$ the function has local maximum value is $f(1)=1$

Example 11

Find the points of local maximum and local minimum of the function

$f: f(x) = \begin{cases} x+1 & , x \le -1 \\ x^2-1 & , x>-1 \end{cases}$

Solution

$\because f$ is continuous at $x=-1$, $f(-1)=0$

, $\because f'(x)=1$ when $x<-1$, $f'(x)=2x$ when $x>-1$

Discussing differentiability at $x=-1$ we find that :

$f'(-1)=1$, $f'(-1^+)=-2$ $\therefore f'(-1^-) \neq f'(-1^+)$

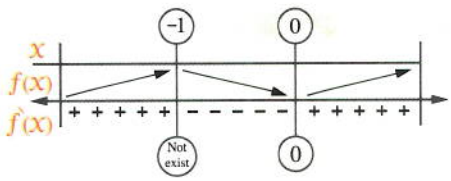
$\therefore f'(-1)$ is not exist.

$\because f'(x) = \begin{cases} 1 & , x<-1 \\ \text{not exist} & , x=-1 \\ 2x & , x>-1 \end{cases}$

$\because f'(x)=0$ at $2x=0$ $\therefore x=0$ (satisfies $x>-1$)

, $f'(x)$ is not exist at $x=-1$

$\therefore$ There are a critical points at $x=0$, $x=-1$



At $x=0$ there is a local minimum value , $f(0)=-1$ $\therefore$ The local minimum point $(0,-1)$

At $x=-1$ there is a local maximum value , $f(-1)=0$ $\therefore$ The local maximum point $(-1,0)$

Example 12

Find the points of the local maximum and local minimum of the function
$f : f(x) = |x - 1|$

Solution

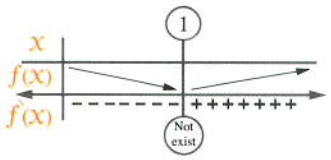
$\because f(x) = |x-1| = \begin{cases} x-1 & , \ x \geq 1 \\ -x+1 & , \ x < 1 \end{cases}$

$\because f$ is continuous function at $x = 1$, $f(1) = 0$

$\because \acute{f}(x) = 1$, when $x > 1$, $\acute{f}(x) = -1$ when $x < 1$

$\acute{f}(1^+) = 1$, $\acute{f}(1^-) = -1$ $\therefore \acute{f}(1)$ does not exist.

$$\therefore \acute{f}(x) = \begin{cases} 1 & , \ x > 1 \\ \text{not exist} & , \ x = 1 \\ -1 & , \ x < 1 \end{cases}$$



$\therefore$ At $x = 1$, there is a local minimum , $f(1) = 0$ $\therefore$ The local minimum point = $(1 , 0)$

Example 13

Find the local maximum and local minimum values of the function

$f : f(x) = \sqrt{9 - x^2}$

Solution

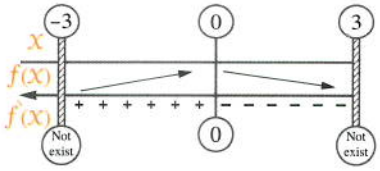
$\because f(x) = \sqrt{9-x^2}$

$\therefore$ The domain of the function is $[-3 , 3]$

$\because \acute{f}(x) = \dfrac{-2x}{2\sqrt{9-x^2}} = \dfrac{-x}{\sqrt{9-x^2}}$

Put $\acute{f}(x) = 0$ $\therefore x = 0$

$\acute{f}(x)$ is undefined at $x = 3$, $x = -3$



$\therefore$ At $x = 0$ there is a local maximum value
and the maximum value = $f(0) = 3$

, at $x = -3$ there is a critical terminal point $(-3 , 0)$

, at $x = 3$ there is a critical terminal point $(3 , 0)$

Notice that

There is an argument between books and resources to consider the terminal points maximum or minimum as most commonly to study the points of local maximum or minimum on an open interval and so in the example at $x = -3$ there is a critical terminal point and at $x = 3$, there is a critical terminal point and we will not call them local maximum or minimum but in the next lesson. we will call them absolute maximum / minimum value.

Remark

If the function f is a polynomial of nth degree , then it has $(n-1)$ of local maximum or local minimum value at most.

For example :

If $f(x) = x^3 + 5x^2 + 6x$ (of 3rd degree) , then $f(x)$ has two values at most of local maximum or minimum.

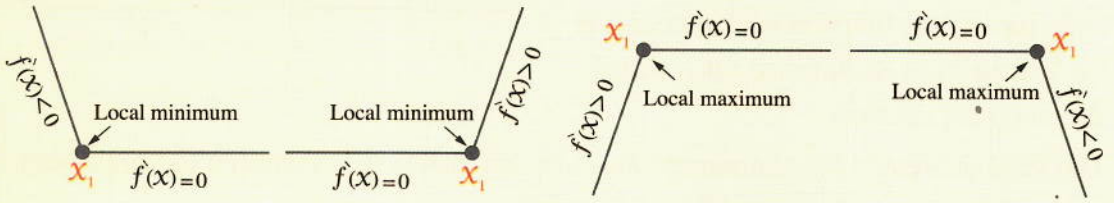
Enrich your knowledge

⊙ When studying the local maximum and minimum values of a defined function on a part of its domain as a constant function , and the x-coordinate of the critical point is x_1 , then :

1. $f'(x) \geq 0$ on the left of x_1 , $f'(x) \leq 0$ on the right of x_1 , **i.e.** $f'(x)$ changes its sign around x_1 from [positive or (zero)] to [negative or (zero)] → then → $(x_1, f(x_1))$ is a point of local maximum value

2. $f'(x) \leq 0$ on the left of x_1 , $f'(x) \geq 0$ on the right of x_1 **i.e.** $f'(x)$ changes its sign around x_1 from [negative or (zero)] to [positive or (zero)] → then → $(x_1, f(x_1))$ is a point of local minimum value

and the following figures illustrate that



⊙ If a continuous function is constant on an interval $]a, b[$, then all points $x_1 \in]a, b[$ are considered critical points because $f'(x_1) = 0$ and so $f(x_1)$ could be local maximum and in the same time local minimum too.